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Challenges and strategies for fast charge of Li-ion batteries

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Abstract: Long charging time for Li-ion batteries is a critical obstacle for the widespread adoption of electric vehicles especially when compared with the rapid refueling of traditional internal combustion engine vehicles. Fast charging results in accelerated performance degradation and low energy efficiency for Li-ion batteries. The batteries adopted in the present electric vehicle market are mainly based on chemistry consisting of a graphite anode and a layered lithium transition metal oxide cathode. The fast-charging capability of such batteries is limited by Li plating on the graphite anode and structural instability of the layered lithium transition metal oxide cathode. In this article, the challenges and possible solutions to these fast charge problems are reviewed at both the materials and cell levels.

1. Introduction

Fast charge is a critical challenge for the widespread adoption of the battery-powered electric vehicles (EVs) compared with the rapid refueling of the traditional internal combustion engine vehicles.^[1] The batteries used in the present EV market are dominated by the chemistry consisting of a graphite anode, a layered lithium transition metal oxide cathode (hereafter referred to as "layered cathode"), and a nonaqueous electrolyte. Such batteries are unable to accept charge rapidly without a detrimental impact on performance and safety. At the cell level, fast charge results in accelerated performance degradation and a rapid temperature rise, which is linked to a number of physical and chemical processes occurring inside the battery. With a focus on the battery chemistry consisting of a graphite anode and a layered cathode, in this article the origins for performance degradation and the possible solutions to these fast charge problems are

reviewed by starting with the fundamental impedance characteristics of Li-ion batteries.

2. Impedance of Li-ion batteries

Ac-impedance spectra of Li-ion batteries are typically composed of two over-lapping semicircles and a straight sloping line. Spectra can be fit by an equivalent circuit as shown in Figure 1a. According to the timescale of physical and chemical processes, the intercept in the beginning is attributed to bulk resistance (R_b), the diameter of the first semicircle in high frequencies to surface layer resistance (R_{sl}), that of the second semicircle in medium frequencies to charge-transfer resistance (R_{ct}), and the straight sloping line in the low frequency end to Warburg impedance (Z_w). Contributed by all cell components, the R_b measures the electronic conductivity of the electrodes, current collectors and tabs, and the ionic conductivity of the electrolyte absorbed throughout the battery; The R_{sl} corresponds to ohmic resistance of the surface layer or so-called solid electrolyte interphase (SEI) and cathode electrolyte interphase (CEI) on the electrodes; The R_{ct} reflects Faradic resistance that measures the kinetics of electrode reactions occurring on two electrodes; Z_w describes the semi-infinite linear diffusion of Li^+ ions on the electrolyte-electrode interfaces. As the one is much larger than the other, two semicircles of the R_{sl} and R_{ct} are merged together, and in such a case the diameter of the merged semicircle is collectively called cell reaction resistance (R_{cr}).^[2] In response to the state-of-charge (SOC), the R_{ct} changes vastly while the R_b and R_{sl} remain nearly unchanged (Figure 1b), in which the nearly constant R_{sl} is attributed to the fact that the R_{sl} of the anode and cathode are changed inversely with the SOC and their values are canceled each other out.^[3]

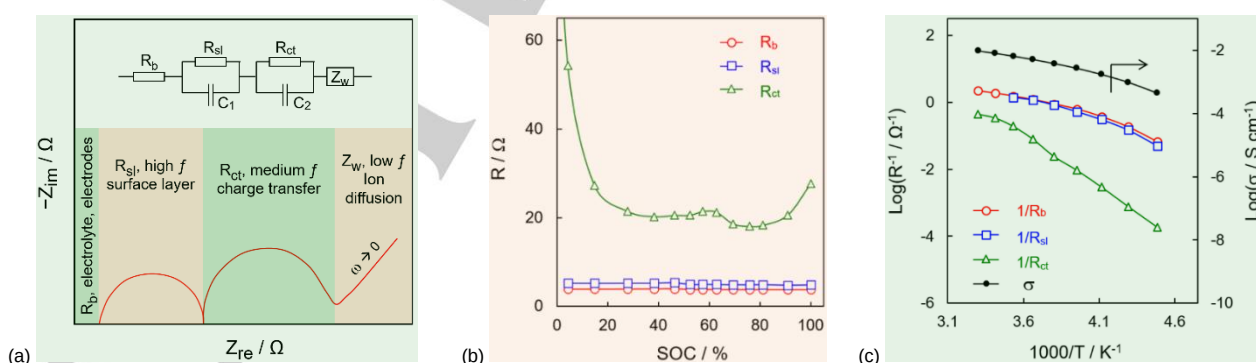


Figure 1. (a) A typical impedance spectrum of Li-ion batteries, (b) SOC-dependence of the R_b , R_{sl} , and R_{ct} , and (c) Arrhenius plot of the electrolyte conductivity and the reciprocal of the R_b , R_{sl} , and R_{ct} measured at 3.87 V (~70% SOC). Figure 1b and Figure 1c are based on a Gr/NCA Li-ion cell with a 1.0 m LiPF_6 3:7 (wt) EC/EMC electrolyte, and replotted from the data of Reference [3].

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Although their values change from cell to cell, the R_b , R_{sl} , and R_{ct} present a very consistent trends in their activation energy. As indicated in Figure 1c, Arrhenius plots of the $1/R_b$ and $1/R_{sl}$ are very similar to that of the electrolyte conductivity (σ) in the pattern (i.e., a smooth curve), being a typical behavior of the ion conduction in a liquid or an amorphous phase with low activation energy. Whereas the Arrhenius plot of the $1/R_{ct}$ is a straight line with a large slope (i.e., high activation energy), indicating a typical behavior of the electrochemical processes. The above differences will help in understanding the problems with fast charge in Li-ion batteries.

3. Challenges and strategies

Performance degradation manifests as a decay in the specific capacity and operating voltage of a cell. This is due to a number of physical and chemical processes occurring inside the battery during fast charge. Physically, Joule heat generated by the high currents causes a rapid rise in temperature, which aggravates parasitic reactions between the electrodes and electrolyte. Chemically, fast charge results in a decrease in coulombic efficiency, and is always accompanied by increases in impedance and volume of the battery, and depletion of the electrolyte. The performance degradation can be ultimately attributed to the loss in cyclable Li^+ ions and the resultant growth in the battery impedance, as schematically summarized in Figure 2 and to be in-depth discussed in the next subsections.

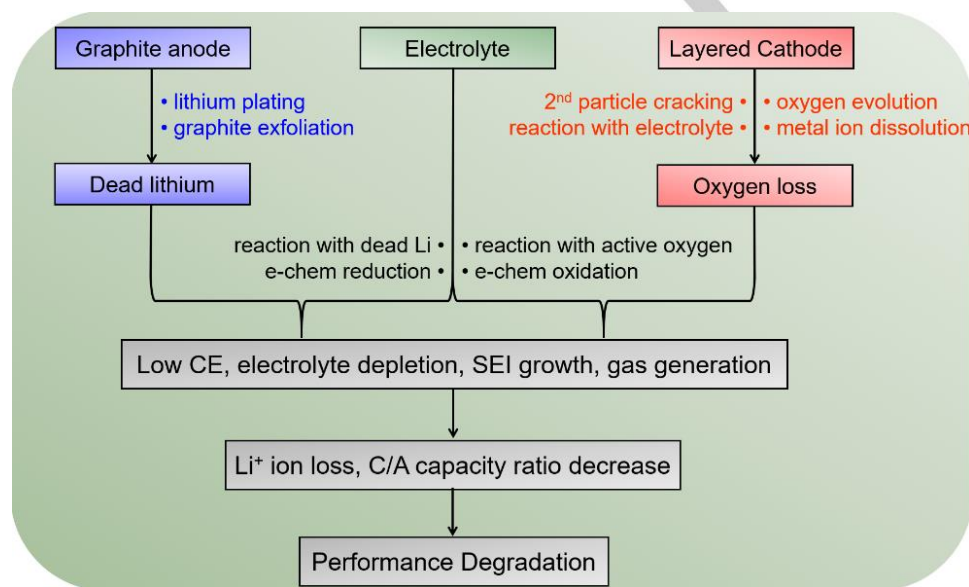


Figure 2. A schematic summary of the mechanisms for performance degradation of Li-ion batteries during fast charge.

3.1. Graphite anode

Li plating is thought to be the primary origin for performance degradation, which often occurs when the batteries are charged at low temperatures^[4] and high rates,^[5] especially in high energy batteries^[6] and in high SOC regions.^[5b, 7] Due to the high reactivity of Li metal, the plated Li reacts with electrolyte solvents, which not only reduces coulombic efficiency of the charging process but also depletes electrolyte leading to impedance growth and volumetric swelling of the battery.^[8] On the other hand, the plated Li can chemically re-intercalate into graphite via micro-galvanic cells as long as they are in contact with graphite and in the presence of electrolyte.^[9] Therefore, there are no reliable techniques that can quantitatively measure Li plating unless a reference electrode is embedded into the battery to record real-time potential of the graphite anode.^[9a, 10] All Li observed in the post-mortem analyses are either isolated (dead) Li or those in the regions where the electrolyte is completely depleted. The results obtained from non-destructive methods, such as those by

analyzing the change in voltage or differential voltage during post-charging relaxation,^[9d, 11] and those by observing the change in capacity or differential capacity in the subsequent discharge,^[12] are changed with rest time due to the issue that the plated Li chemically re-intercalates into graphite.

Theoretically, Li plating occurs and competes with the normal intercalation of Li^+ ions into graphite as long as the potential of the anode falls below 0 V vs. Li/Li^+ . The intercalation of Li^+ ions into graphite occurs mainly in a narrow potential range of 0.005–0.300 V vs. Li/Li^+ , and such low potentials can be easily achieved by the voltage polarization during fast charge. Therefore, lowering the voltage polarization is the only solution to mitigating Li plating. In this regard, reducing the thickness (accordingly the loading) and tortuosity of graphite anode is shown to be effective in facilitating ion transport in the electrode, and hence mitigate Li plating.^[7b, 7c] The graphite anode with low tortuosity, such as those made by laser cutting^[9d] and freeze-casting,^[13] is shown to offer much improved rate capability and reduced Li plating. Another strategy is to enhance the kinetics of the electrode reactions by raising the temperature of the charging process. As shown in

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Figure 1c, the activation energy of the $1/R_{ct}$ is inherently large compared with those of the $1/R_b$ and $1/R_{sl}$, suggesting that the polarization resulting from R_{ct} can be dramatically reduced by raising the temperature. As such, Yang et al.^[11] proposed charging the battery at $\sim 60^\circ\text{C}$ and discharging at ambient temperature, finding that a high energy pouch cell (209 Wh kg^{-1} , Gr/NMC532) could retain 91.7% of capacity after 2,500 cycles at a 10 min charging rate. In addition, chemical heteroatom doping^[14] and physical hybridizing with other anode materials that have higher lithiation potentials or better rate capability compared with the pristine graphite, such as hard carbon,^[15] amorphous Si,^[16] and SiO,^[17] are shown to some extent to alleviate Li plating.

The impact of Li plating on safety during fast charge seems to be overestimated. The safety risk in batteries is determined by two factors, the energy stored (namely SOC) and the temperature. First, most batteries cannot be charged to 100% SOC or sent to an overcharged state by fast charge. Instead, the cathode must be undercharged as long as Li plating occurs because of the " $U = E_+ - E_-$ " correlation between the cell voltage (U) and the potentials of the cathode (E_+) and anode (E_-), as shown in Figure 3; Second, the temperature rise by Joule heating is unlikely to be so large as to trigger thermal runaway as long as the battery thermal management system functions appropriately; Third, the chance to form Li dendrites that can pierce the separator is low because the plated Li chemically re-intercalates into the graphite. Instead, the severe issue with Li plating is the chemical reactions between the plated Li and electrolyte solvents, which not only consumes the cyclable Li^+ ions and electrolyte but also results in increases in the battery's impedance, volume (swelling) and electrode's thickness.^[18] Furthermore, the "dead Li" formed by the reactions greatly lowers the thermal runaway temperature, which could partially explain why the majority of EV fire incidents (81% of 79 incidents) were initiated during parking and driving, rather than during the charging process, after the EVs had served for 1–2 years (being an issue of the battery aging due to accumulation of dead Li).^[19] In addition, the reactions cause inhomogeneity in the distribution of electrolyte, and in the resistance and thickness of the electrode.

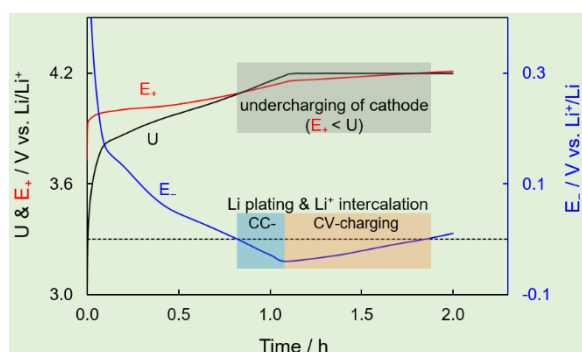


Figure 3. The correlation between cell voltage and potentials of the cathode and anode during the standard CCCV-charging process, showing that Li plating is accompanied by the undercharging of the cathode. That is, the potential of the cathode (E_+) must be lower in the value than the cell's voltage (U) when Li plating occurs.

Although Li plating has been overwhelmingly considered the culprit of fast charge problems, it was found^[10c, 20] that even if the potential of the graphite anode never drops below 0 V vs. Li/Li^+ during 10C-charging in power-optimized Gr/NCA cells, the accelerated performance degradation is still unavoidable. Similarly, Mussa et al.^[21] reported that energy-optimized batteries (25 Ah, Gr/NMC111) release large amounts of gas, accompanied by severe graphite exfoliation, as the batteries are charged at 4C. The above results point to the fact that under fast charge, the existing SEI on the graphite anode is unable to protect electrolyte solvents from electrochemical reduction. This is yet another issue for fast charge resulting in electrolyte depletion, impedance growth, and battery swelling.

3.2. Layered cathode

Numerous factors lead to performance degradation of the layered cathode, including alkaline Li residual compounds (mainly Li_2CO_3 , LiOH , and Li_2O) on the particle surface, Li-M cationic mixing, oxygen evolution, irreversible layered-spinel-rocksalt phase transformation, transition metal ion dissolution, fracture of the secondary particle structure, and parasitic reactions with the electrolyte.^[22] Except for the alkaline Li residual compounds that cause slurry gelation in the coating process and generate gas in the initial cycles, all other factors can be ultimately attributed to the loss of oxygen.^[22f] The lost oxygen is either released in the form of molecular oxygen or reacts with the electrolyte solvents to form solid Li compounds (such as Li_2O , Li_2CO_3 , and organic Li salts) and gas products (such as CO_2 and CO).^[23] As a result, both the active material and the cyclable Li^+ ions are lost, leading to a decrease in the cathode-to-anode (C/A) capacity ratio, accompanied by electrolyte depletion, impedance growth, and volumetric swelling.

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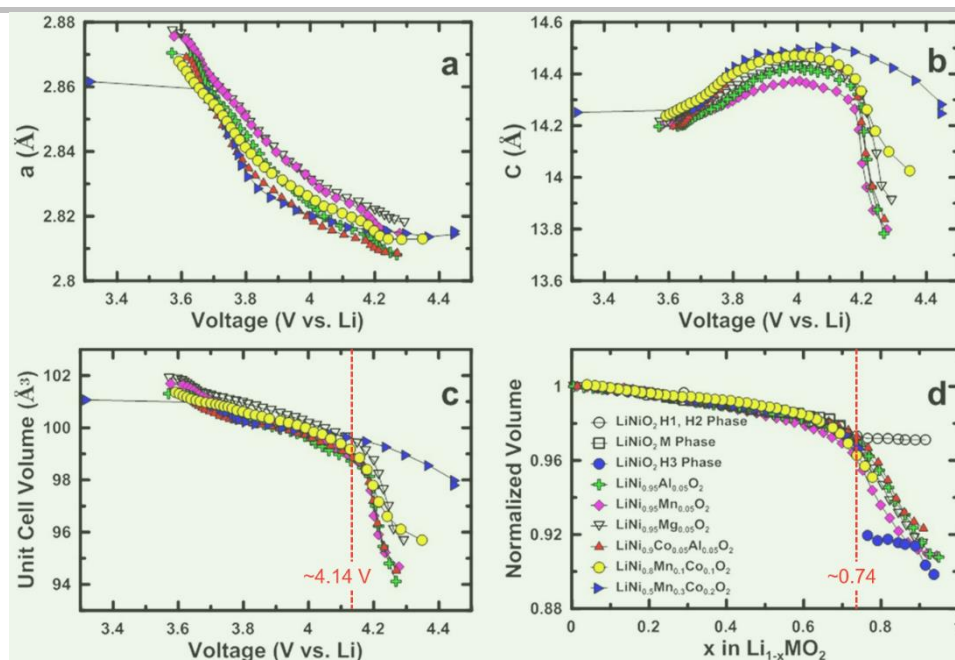


Figure 4. A-axis lattice parameter (a), c-axis lattice parameter (b), unit cell volume (c), and normalized unit cell volume (d) as a function of voltage or x in various layered Li_xMO_2 compounds, i.e., $\text{LiNi}_{0.95}\text{Al}_{0.05}\text{O}_2$, $\text{LiNi}_{0.95}\text{Mn}_{0.05}\text{O}_2$, $\text{LiNi}_{0.95}\text{Mg}_{0.05}\text{O}_2$, $\text{LiNi}_{0.9}\text{Co}_{0.05}\text{Al}_{0.05}\text{O}_2$, $\text{LiNi}_{0.8}\text{Mn}_{0.1}\text{Co}_{0.1}\text{O}_2$, and $\text{LiNi}_{0.5}\text{Mn}_{0.3}\text{Co}_{0.2}\text{O}_2$. Reprinted from Reference [22d] Copyright (2019) with permission from the American Chemical Society.

The most critical challenge of the layered cathode materials is their structural instability against rapid changes in the unit cell volume and mechanical stress during fast charge. The layered cathode materials are typically present in a spherical secondary structure composed of numerous aggregates of nanosized primary particles. Such secondary structures are unable to withstand rapid changes in the volume and mechanical stress, especially in the regions where the potential is higher than 4.14 V vs. Li/Li^+ or the SOC is higher than ~74% as shown in Figure 4.^[22d] As a consequence, microcracks are developed throughout the spherical particle, creating new surfaces on the walls of microcracks, where the electrolyte will permeate. The resultant reactions between the electrolyte solvents and active materials in the newly created cracks lead to detrimental SEI growth and gas generation. Taking structural instability of the spherical secondary structure and large surface area of the nanosized primary particles into consideration, single-crystal materials are in principle beneficial to stabilizing the performance of the layered cathode. For example, Harlow et al.^[24] demonstrated that in a wide temperature range (20–55 °C), the Li-ion cells employing single-crystal NMC532 provide far longer lifetimes when compared with cells using spherical aggregates. The increased lifetime was attributed to two properties of the single-crystal materials: (1) intrinsically better stability of the single-crystal primary structure compared to the spherical secondary structure, and (2) significantly smaller specific surface area of the large single-crystals compared to the nanosized primary particles.

Another issue with the layered cathode materials is the sluggish mobility of Li^+ ions. It was determined that the diffusion coefficient of Li^+ ions in NCA is $\sim 10^{-10} \text{ cm}^2 \text{ s}^{-1}$,^[25] while that in graphite is as high as 10^{-7} – $10^{-6} \text{ cm}^2 \text{ s}^{-1}$.^[26] In other words, the fast-charging capability of Li-ion batteries is kinetically limited by the layered cathode, rather than the graphite anode. This

conclusion has been verified by the impedance analyses using three-electrode Li-ion cells^[10a, 10c] and symmetric cells,^[7b] respectively, indicating that the graphite anode has significantly smaller impedance than the cathode. With cycling, however, the impedance of the graphite anode grows much faster than that of the cathode, ascribed to Li plating on the graphite anode and detrimental participation of the transition metal ions dissolved from the layered cathode into the existing SEI.^[10a] Although the sluggish mobility of Li^+ ions is an intrinsic property of the layered cathode materials, the kinetics of the electrode reactions can be improved by reducing the path of Li^+ ion transport with electrode engineering. Therefore, strategies such as reducing the particle size of the layered cathode materials^[27] and reducing the thickness^[28] and tortuosity^[9d, 13] of the electrodes are commonly used to enhance the fast-charging capability. However, these strategies unavoidably result in a trade-off in capacity and cost.

3.3. Electrolyte

Electrolyte enables electrode reactions by transporting Li^+ ions between the electrodes and within the electrodes. Since Li^+ ions are more mobile in liquids than in solids, ionic conductivity of the liquid electrolytes is unlikely a rate-limiting factor for fast charge unless the loadings (accordingly the thicknesses) of the electrodes are extremely high. However, the electrolyte vitally affects Li plating and rate capability via ohmic polarization (R_b and R_s) and electrode reaction kinetics (R_{ct}). The values of R_b and R_s in principle follow ionic conductivity of the electrolyte, whereas that of the R_{ct} is impacted not only by ionic conductivity of the electrolyte but also by the activation energy of the Li^+ ion solvation and desolvation in the electrode reactions. Employing an electrolyte with high ionic conductivity is always the first choice for enhancing the fast-charging capability of Li-ion batteries. The

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strategies for increasing ionic conductivity of the electrolytes are either or both modifying the existing solvent blend and using an alternative Li salt with a high dissociating ability.^[8, 29] The solvent blends are commonly modified by introducing a co-solvent that has the combined merits of viscosity, polarity (dielectric constant), and boiling point. Suitable co-solvents are, but are not limited to, ethers,^[30] esters,^[30a, 31] and nitriles.^[32] For ethers, Zhang^[30a] increased ionic conductivity of a 1.0 M LiPF₆ 3:7 (wt) EC-EMC electrolyte from 8.6 S cm⁻¹ to 11.3 S cm⁻¹ (20 °C) by replacing 10 wt% EMC with DME; With esters, Logan et al.^[31d] raised ionic conductivity of a 1.0 M LiPF₆ 3:7 (wt) EC-DMC electrolyte from 11.14 S cm⁻¹ to 13.37 S cm⁻¹ (20 °C) by replacing 20 wt% DMC with MA; For nitriles, Cho et al.^[32] enhanced ionic conductivity of a 1.0 M LiPF₆ 1:1 (vol) EC-DMC electrolyte from 11.33 mS cm⁻¹ to 16.73 mS cm⁻¹ (25 °C) by reformulating it as a 1.0 M LiPF₆ 1:1:1 (vol) EC-DMC-PN electrolyte. All the above modifications in the solvent blend were proven to enhance the rate capability of the Li-ion batteries. For Li salts, LiFSI was considered to be a magic salt because of its unique ability in dissociating and facilitating the formation of a LiF-rich SEI,^[33] although anodic corrosion of the Al current collector at high potentials might still be a problematic issue.^[34] For example, Du et al.^[35] realized a 13% improvement in the capacity of a 60 Ah Gr/NMC811 pouch cell in a fast-charging test (5C, 12 min) by replacing LiPF₆ with LiFSI in a 1.5 M LiPF₆ 3:7 (wt) EC-EMC electrolyte. They contributed the improvement to increased ionic conductivity (from 9.35 to 10.36 mS cm⁻¹ at 30 °C) and Li⁺ transference number (from 0.382 to 0.495).

The activation energy of the Li⁺ ion solvation and desolvation is another factor decisively affecting the fast-charging capability. In the charging process, Li⁺ ions are solvated and desolvated simultaneously on the cathode and anode, respectively, in addition to the deintercalation and intercalation reactions. Several studies have unveiled that the Li⁺ ion solvation and desolvation activation energies play an even more important role than the ionic conductivity in affecting the cycling performance of Li-ion batteries at high rates and low temperatures. For example, Zhang et al.^[36] showed that at low temperatures (-20 and -30 °C), Gr/NCA cells deliver higher capacity with a LiBF₄ electrolyte than with a LiPF₆ counterpart while the LiBF₄ electrolyte has lower ionic conductivity. Similarly, Yamada et al.^[37] demonstrated that a Li/Gr cell with 3.6 M LiFSI-DME electrolyte having a lower ionic conductivity (7.2 mS cm⁻¹) outperforms that with 1.0 M LiPF₆-EC-DMC electrolyte (~10 mS cm⁻¹); Zhang et al.^[30b] reported that at 5C, a Gr/NMC811 cell with 2.8 mAh cm⁻² of NMC811 loading demonstrates higher capacity with a 1.4 M LiFSI DMC-EC-TTE electrolyte (2:0.2:3 by mol, 1.07 mS cm⁻¹) than with a 1.0 M LiPF₆ EC-EMC electrolyte (3:7 by wt + 2 wt% VC, 6.07 mS cm⁻¹). The above results reveal that the ionic conductivity of the electrolytes is not the only factor governing the fast-charge capability. Although poorly understood, the importance of Li⁺ solvation and desolvation activation energy in fast-charge capability cannot be overlooked.

As suggested by their similar values, the R_b and R_{sl} nearly contribute equally to the ohmic polarization (see Figure 1b). Not impacting the bulk electrolyte conductivity, electrolyte additives are shown to be very effective in reducing the R_{sl}. Electrolyte additives enhance the rate capability of batteries by two means, (1) constituting SEI with high ionic conductivity, and (2) suppressing detrimental SEI growth. FI is an excellent example of the first function.^[38] It is shown that adding 2 wt% FI into a 1.0 M LiPF₆ 1:1 (vol) EC-DMC electrolyte facilitates the formation of a

slightly thicker but more conductive SEI, which features an inorganic-rich inner layer and an organic-rich outer layer. Such an SEI has proven to dramatically improve the rate capability of the Li/Gr cell at both room temperature (20 °C) and low temperatures (0 and -20 °C). Examples for the second function include, but are not limited to, a variety of organic compounds such as VC,^[39] PS,^[39b] FEC,^[39b, 40] and TPP,^[41] and Li salts such as LiFSI,^[42] LiPO₂F₂,^[43] LiDFBOP.^[44] As an example, the SEI formed with a 0.5 wt% LiPO₂F₂ additive effectively mitigates parasitic reactions between the electrodes and electrolyte in a Gr/NMC532 pouch cell, and consequently the cell shows higher coulombic efficiency, longer cycle life, and much better rate capability, which is ascribed to the significantly suppressed impedance growth.^[43] For the layered cathode, most of the improvements by using electrolyte additives are based on reactions with the surficial Li residual compounds (such as Li₂CO₃, LiOH, and Li₂O) to reconstitute a more conductive or more favorable CEI.^[45] Therefore, effective additives for the layered cathode are mainly centered on the compounds that can react with the alkaline Li residual compounds, such as sulfates & sultones,^[46] phosphates & phosphites,^[46b, 47] and borates.^[46b, 48] As pointed out in Section 3.1, electrolyte solvents are subject to electrochemical reduction on the graphite anode under fast charge.^[10c, 20] In this view, the electrolyte additives are of particular significance in mitigating capacity fade and stabilizing coulombic efficiency under fast charge. It should be noted that the solvents and Li salts mainly determine the bulk properties (such as ionic conductivity and Li⁺ ion solvation and desolvation activation energy) of the electrolytes, whereas the electrolyte additives affect the electrolyte-electrode interfaces. Success in developing electrolytes for the fast charge of Li-ion batteries depends greatly on a synergistic combination of solvents, Li salts, and electrolyte additives.

3.4. Battery engineering

In addition to the electrode and electrolyte materials, battery engineering impacts fast-charge capability at all levels from materials to electrodes and cells. The material level includes the size and shape of electrode active materials; the electrode level includes the composition, thickness/loading, porosity/tortuosity and the uniformity of the electrodes, as well as the number and position of the tabs; the cell level includes the C/A capacity ratio, the pressure and pressure distribution, and the type and size of the cells. With respect to the requirements for specific applications, the above parameters should be properly coordinated in terms of energy density, power capability, lifetime, and cost. In general, small particle size reduces the path for ion transport, but, it also reduces the materials' tap density while increasing the reactivity with the electrolyte due to the large surface area of the small particles; thin electrode (corresponding to low loading) favors enhanced rate capability, but it reduces the capacity and increases the cost due to the need for more inactive materials including electrolyte, separator, and current collectors; low C/A capacity ratio is beneficial to mitigate Li plating by elevating the operation potential of the graphite anode, but it increases the degradation rate of the cathode and electrolyte due to the exposure to higher SOC for the cathode and higher potentials for the electrolyte.

Heat generation is one of the most severe issues leading to performance degradation during fast charge although a rise in the temperature favors increasing ion transport and electrode

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reaction kinetics. Heat generation originates from two sources, reversible entropic heat and irreversible Joule heat. Under fast charge, Joule heat can constitute over 90% of the total heat,^[49] and it is the major form of energy loss during fast charge. Joule heat equals the total ohmic loss from R_b , R_{sl} and R_{ct} , and it is affected by the current and age of the battery. In view of the battery design, the inhomogeneous distribution of the heat generation is a key issue resulting in accelerated degradation of the electrode and electrolyte materials. A critical challenge for battery engineering is how to homogenize the thickness/loading and porosity/tortuosity of the electrodes, and mechanical pressure between the electrodes throughout the entire battery. All these parameters affect the distribution of the current, and consequently the uniformity of heat generation (hence the temperature), and the degradation rate of the electrode and electrolyte materials. Besides, separator play important roles in homogenizing the current density via its pore distribution, affecting the rate capability via its porosity, pore tortuosity, and contact resistance with the electrodes, and increasing the fast charge safety via its dimensional stability at high temperatures. For example, surface modification with a gel polymer electrolyte reduces the electrode-separator contact resistance, and that with inorganic ceramic powder increases not only the wettability with liquid electrolytes but also the dimensional integrity at high temperatures.^[50]

Heat dissipation is particularly important for high energy and large batteries. Even for a 18650 cell (3 Ah, NCA), its peak temperature in the electrode jellyroll can reach 60 °C when being charged at 2.2 C (6.6 A).^[51] For pouch cells^[52] and prismatic cells,^[53] temperatures close to the tabs and the core are higher than other regions due to the higher current density near the tabs and poor heat dissipation in the core. The temperature gradient leads to inhomogeneity in the battery internal resistance, which consequently aggravates inhomogeneity in the current, heat generation, and materials' degradation rate. On the other hand, the number and position of tabs play an important role in homogenizing the current distribution, and hence affect heat generation and temperature-related degradation.^[54] For mechanical pressure, two inverse impacts on the rate capability must be taken into account. Increased pressure reduces contact resistance, however, it also squeezes electrolyte out of the separator. For example, a difference in the pressure caused by a label on the pouch cell cover can cause a large difference in Li plating (and hence in the thickness and impedance of the graphite anode) when cells with an 11.7 mg/cm² graphite loading were cycled by charging at 6C under 4 psi pressure.^[55]

3.5. Charging protocols

At present, the industrial standard charging protocol for Li-ion batteries is the constant current-constant voltage (CCCV) scheme. Aiming to charge batteries quickly while maintaining health, a number of alternative charging protocols have been proposed.^[1b, 1c] However, none of these protocols takes the specific chemistry of cells into account. A fast-charge test on a group of Panasonic NCR 18650B cells (3.2 Ah, NCA)^[56] showed that the capacity faded quickly and their cases were broken after 60 cycles between 2.5 V and 4.2 V by the CCCV charging at 2C. Post-mortem analyses found that numerous cracks and solid white residues formed on both the graphite anode and NCA cathode, as shown in Figure 5. In particular, the white solid residues on the

graphite anode were identified by XRD to be MCO_3 , MF_2 ($\text{M}=\text{Ni}$ and Co), LiOH , Al-Li alloys, and other XRD unidentifiable organic polymers, indicating that severe dissolution of transition metal ions from the NCA cathode occurred during the 2C CCCV cycling. Also, the M^{2+} ion dissolution (in relation to the loss of oxygen^[22f]) and resulting structural degradation of NCA were independently observed from Gr/NCA coin cells after 200 cycles by 10C CCCV charging.^[20] In fact, the severest safety risk for fast charge is the heat and gas generation, which triggers cell venting, so that hot flammable gas and electrolyte vapor release and catch fire once they meet an ignition source.

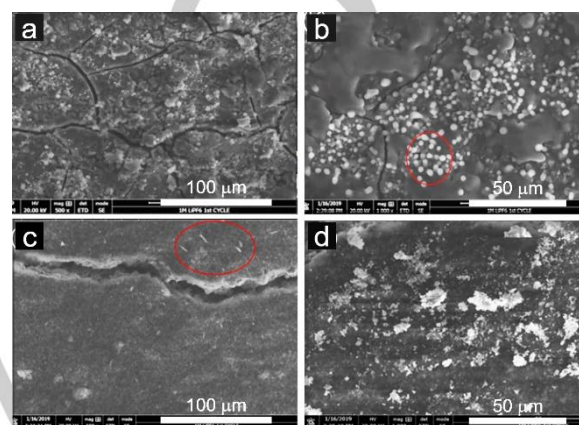


Figure 5. SEM images of the anode (a, b), and the cathode (c & d) of a Panasonic NCR 18650B batteries after 60 CCCV-charging cycles at 2C between 2.5 V and 4.2 V. Reprinted from Reference^[56] Copyright (2020) with permission from John Wiley & Sons, Ltd.

The capacity fade in Li-ion batteries under fast charge typically shows a rapid loss in early cycling, followed by a decelerated rate in later cycles.^[4b, 20, 56-57] Interestingly, the rapid capacity loss is always accompanied by low coulombic efficiencies even without Li plating, likely caused by electrochemical reduction of the electrolyte solvents.^[20, 57b] A good charging protocol should account for the chemistry of the batteries, and be able to smartly adjust current with the SOC. As shown in Figure 1b, the R_{ct} in low SOC regions ($< \sim 20\%$, this number varies with the chemistry and C/A capacity ratio of the cells) is much larger than those in the other SOC regions. In order to limit Joule heat and voltage polarization, charging rate in the low SOC regions should be relatively low. In this regard, Sebastian et al.^[56] proposed an internal resistance based fast charging protocol. They cycled a group of Panasonic NCR 18650B batteries (3.2 Ah, NCA) by first charging at 1C until the internal resistance declines to a certain level and then charging at 2C CCCV for the remainder. As a result, both capacity retention and mechanical stability of the batteries were greatly improved while the total charging times remained similar to those using the conventional 2C CCCV charging. Besides, Sieg et al.^[10d] proposed a fast-charging protocol that allows the maximum charging rate without Li plating by inserting a constant-potential charging step at 0 V vs. Li/Li^+ of the graphite anode into the standard CCCV protocol. By embedding a Li reference electrode into the cell, they were able to first CC-charge the cell until the potential of the graphite anode reached 0 V vs. Li/Li^+ , then maintained the potential of the graphite anode at 0 V vs Li/Li^+ until the cell's voltage reached the

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upper limit, and finally CV-charged until the end. Although embedding a Li reference electrode is not practical for the commercial Li-ion batteries, this charging pattern provides a good example for the design of charging protocol to take the chemistry of the cells into account.

One the other hand, as shown in Figure 4d, a dramatic decrease in the unit cell volume of the layered cathode materials starts at ~74% SOC, above which the layered cathode materials suffer from severe volume shrinkage and mechanical stress, resulting in fracture of the secondary particle structure, oxygen evolution, and transition metal ion dissolution.^[22f] In view of the chemical and structural stability of the layered cathode materials, the charging process should be terminated before SOC reaches 75%. Note that the instability of the layered cathode materials is more related to the SOC (i.e., ~75% SOC in Figure 4d, this number remains nearly constant regardless of the Ni content), rather than the potential (i.e., ~4.14 V in Figure 4c, this number increases with a decrease in the Ni content).^[58] Therefore, the SOC is a better parameter than voltage to be used for terminating the charging process. For example, Zhang^[20] found that the capacity retention of Gr/NCA cells in 10C CCCV-charging cycling can be dramatically stabilized by lowering the charging cutoff voltage to 4.0 V from the normal 4.2 V. Similarly, Yang et al.^[11] demonstrated that high energy Gr/NMC532 pouch cells (9.5 Ah, 170 Wh kg⁻¹) can retain 91.7% of the original capacity after 2,500 cycles at a 10-min charging rate by limiting the capacity to 80% SOC and charging the battery at ~60 °C. The above facts reveal that both the large internal resistance at low SOC and the chemical and structural instability of the layered cathode materials at high SOC should be taken into account for an ideal fast-charging protocol.

4. Summary and outlook

The challenges for the fast charge of Li-ion batteries are outlined and the strategies towards solving the relative problems are reviewed. Briefly, fast charge results in accelerated performance degradation and low energy efficiency. Performance degradation manifests as capacity loss and increased voltage polarization, and it is always accompanied by low coulombic efficiencies. Low energy efficiency is mainly due to the loss from Joule heat, and furthermore, the resultant rise in temperature aggravates chemical degradation of the electrode and electrolyte materials. Low coulombic efficiency results from Li plating on the graphite anode, chemical degradation of the layered cathode materials in high SOC, and electrochemical redox of the electrolyte solvents on the electrodes. The above processes are accompanied by electrolyte depletion, as well as impedance growth and volumetric swelling of the batteries. Poor heat dissipation results in an inhomogeneous distribution of temperature, causing localized chemical degradation of the electrode and electrolyte materials. In most cases, improvements of the fast-charging capability come with trade-offs in other capabilities, such as capacity, lifetime, and cost. At the materials and cell levels, future efforts for enhancing the fast-charging capability can be centered on the following aspects:

1. Graphite anode: The challenge for the graphite anode is Li plating, rather than the electrode reaction kinetics. The primary origin for Li plating is the negative overpotentials vs. the equilibrium potential of Li metal, for which the only solution is to

reduce internal resistance (R_b , R_{sl} , and R_{ct}) of the battery. Heteroatom doping on graphite and optimizing the composition and structure of the graphite electrode would help reduce R_{ct} and R_b . Surface modification of graphite particles impacts the formation and composition of the SEI, which is another direction for the reduction of R_{sl} .

2. Layered cathode: The cathode kinetically limits the fast-charge capability, its challenge is the chemical and structural instability of the layered cathode materials at high SOC. As the SOC exceeds ~75%, the layered cathode materials are subject to dramatic shrinkage in the unit cell volume, leading to oxygen evolution, transition metal ion dissolution, and secondary particle structure cracking. This not only results in the loss of active material but also triggers parasitic reactions with the electrolyte solvents. In principle, physical surface coating and modification would protect the active material from reactions with the electrolyte, and chemical doping with foreign cations stabilizes the primary structure (namely the crystalline structure). The simplest strategy for stabilizing the layered cathode is to avoid high SOC (>75%), this however will compromise capacity.

3. Electrolyte: Internal resistance and electrode reaction kinetics of Li-ion batteries are vitally impacted by the electrolyte. The ionic conductivity and Li⁺ ion solvation-desolvation activation energy vitally affect the fast-charging capability. Meanwhile, the chemical and electrochemical stabilities of the electrolyte solvents on the graphite anode and layered cathode greatly influence coulombic efficiency, aging, and lifetime of the batteries. The above "bulk" properties of the electrolyte can be improved by optimizing solvent blends and Li salts. On the other hand, electrolyte additives "magically" affect the formation and stability of the SEI and CEI, as well as ion transport in the electrolyte-electrode interfaces. Furthermore, the SEI and CEI play essential roles in protecting the active materials from chemical reactions with the electrolyte, and in preventing the electrolyte solvents from electrochemical decomposition on the electrodes. Exploring new electrolyte additives with considering both the anode and cathode is of particular significance for enabling the healthy and safe fast-charge of Li-ion batteries.

In addition to the electrode and electrolyte materials, battery engineering is of equal importance in contributing to fast-charge capability, lifetime, and safety. With respect to specific applications, the fast-charge capability and energy density can be optimized by coordinating the parameters among the composition, thickness/loading, and porosity/tortuosity of the electrodes, the number and position of tabs, and the cost. Uniformity of the current and heat generation is the top priority for guaranteeing healthy and safe fast-charge. Under fast charge, a small variation in the internal resistance will lead to large inhomogeneity in the localized temperature and material degradation. As such, the uniformities of the electrode thickness/loading and porosity/tortuosity, and of the C/A capacity ratio and electrolyte distribution are the prerequisite for enabling fast charge with a long lifetime. In large high energy batteries, heat dissipation crucially affects coulombic efficiency, aging rate, and safety of the batteries. Advanced charging techniques are needed for healthy and safe fast-charge. An ideal charging protocol should be able to dynamically adjust the current rate with the change in SOC, and terminate the charging process before the SOC reaches unstable regions. The scope of this mini-review is limited to the materials and cell levels. For EV applications, the module and pack levels

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necessitate systematic studies by synergistic collaborations between multidisciplinary scientists and engineers.

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Conflict of Interest

The author declares no conflict of interest.

Abbreviation list of chemicals and instruments

DMC	Dimethyl carbonate
DME	1,2-dimethoxyethane
EC	Ethylene carbonate
EMC	Ethyl methyl carbonate
FEC	Fluoroethylene carbonate
FI	Fluorosulfonyl isocyanate
Gr	Graphite
LiDFBOP	Lithium difluorobis(oxalato) phosphate
LiFSI	Lithium bis(fluorosulfonyl)imide
MA	Methyl acetate
MP	Methyl propionate
NCA	$\text{LiNi}_{0.80}\text{Co}_{0.15}\text{Al}_{0.05}\text{O}_2$
NMC532	$\text{LiNi}_{0.50}\text{Mn}_{0.30}\text{Co}_{0.20}\text{O}_2$
NMC811	$\text{LiNi}_{0.80}\text{Mn}_{0.10}\text{Co}_{0.10}\text{O}_2$
PN	Propionitrile
SEM	Scanning electron microscope
TPP	Tripropargyl phosphate
TTE	1,1,2,2-tetrafluoroethyl-2,2,3,3-tetrafluoropropyl ether
VC	Vinylene carbonate
XRD	X-ray diffraction

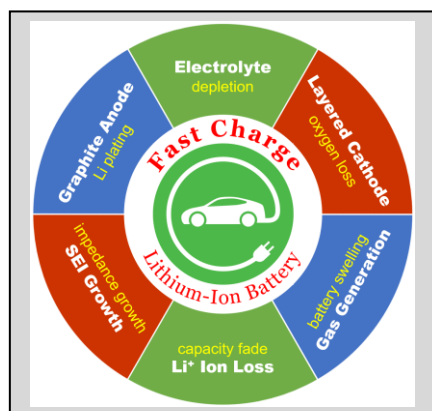
Dr. Sheng S. Zhang (Shengshui Zhang) is a senior research chemist in the U.S. Army Combat Capabilities Development Command (CCDC) Army Research Laboratory (ARL). He has over 30 years experiences in fundamental and applied researches on many electrochemical energy storage devices, including Li metal batteries, Li-air batteries, Li-ion batteries, supercapacitors, and Li-ion capacitors. His current research interests are in both new and existing materials for electrochemical energy storage and their implementation in batteries for Army applications. As an active researcher, he has been recognized by his single- and first-authored publications with a total of 17,700+ citations and an h-index of 66.



Keywords: Fast charge • Li plating • Layered cathode • Performance degradation • Charging protocol.

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Entry for the Table of Contents



Fast charge results in accelerated performance degradation and low energy efficiency, which is primarily attributed to Li plating on graphite anode, structural instability of the layered cathode materials, and electrolyte depletion. Mechanisms and possible solutions to the fast charge problems of Li-ion batteries are critically reviewed.

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